HEALTHCARE COST ANALYSIS

Exploratory Data Analysis & Inferential Statistics

PROJECT REPORT

Prepared from the supplied Healthcare_project.ipynb notebook

Python • pandas • Seaborn • SciPy • statsmodels

Executive Summary

This project investigates medical insurance charges using customer demographic and lifestyle variables. It combines exploratory visualizations with hypothesis tests to assess differences in charges, relationships between variables, and customer distribution across regions. The notebook uses the fields age, sex, bmi, children, smoker, region and charges.

Analysis area	Notebook result	Interpretation at $\alpha = 0.05$
Smoker vs non-smoker charges	Reject H■	Smokers have higher mean charges in the one-sided test.
Charges across regions	Reject H■	At least one regional mean differs.
Gender and smoking status	p = 0.005324	Evidence of association.
BMI by gender	p = 0.089976	Fail to reject H■.
Age and charges	p ≈ 4.89 × 10■²■	Statistically significant linear association.
Regional customer counts	p = 0.324562	Fail to reject H■.
Smoking status and region	p = 0.061720	Fail to reject H■.

A p-value below 0.05 supports rejecting H■ at the 5% significance level. Failing to reject H■ does not prove the null hypothesis true. Statistical association does not establish causation.

1. Project Background

Medical insurance costs can vary according to customer age, body mass index, smoking status, family characteristics and region. This project explores those patterns through visual analysis and hypothesis testing. It is an observational analysis, not a causal study or a predictive model.

2. Objectives

• Describe the distribution of charges.

• Test whether mean charges differ from $13,270.

• Compare charges by gender, smoking status and region.

• Examine gender–smoking and smoking–region associations.

• Compare BMI between genders and assess the age–charges relationship.

• Test whether customers are equally distributed among regions.

3. Dataset and Variables

The notebook reads a file named insurance.csv. The CSV itself was not included with the upload, so this report relies on the notebook's displayed preview and saved outputs.

Variable	Description	Type
age	Age in years	Numerical
sex	Gender category	Categorical
bmi	Body mass index	Numerical
children	Number of children/dependants	Count
smoker	Smoking status (yes/no)	Categorical
region	One of four geographic regions	Categorical
charges	Medical insurance charges	Numerical outcome

Dataset source and license are not specified in the notebook and should be added for formal submission.

4. Tools and Methodology

• pandas and NumPy for data loading and numerical operations.

• Seaborn and Matplotlib for visualizations.

• statsmodels for z-tests, t-tests, ANOVA and categorical association tests.

• SciPy for Pearson correlation testing.

• Significance level: $\alpha = 0.05$.

Workflow: load data → inspect variables → visualize → define H■/H■ → select test → compare p-value with α → report a contextual conclusion.

5. Exploratory Data Analysis

5.1 Distribution of insurance charges

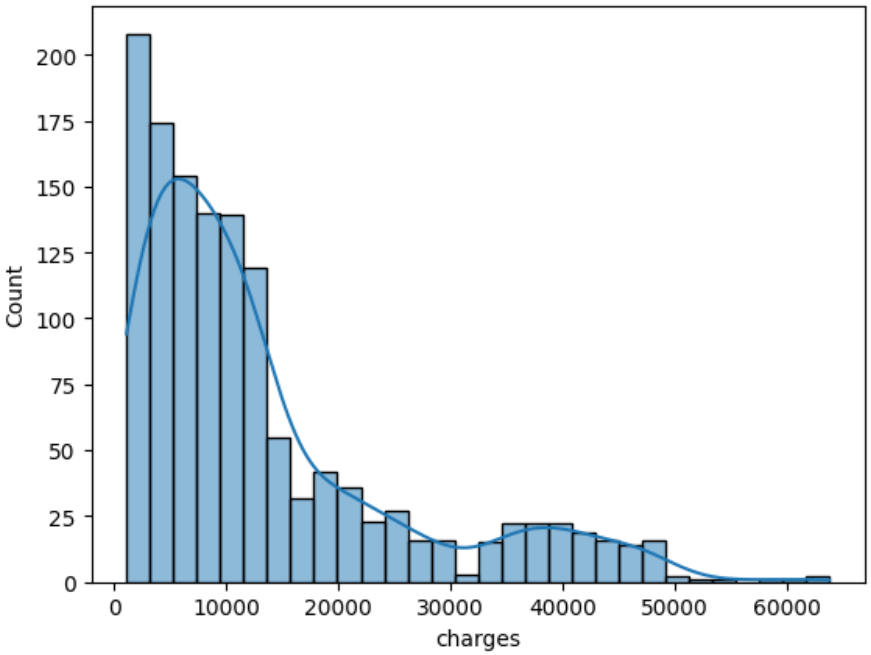


Figure 1. Histogram and KDE of insurance charges.

The histogram and density curve help assess skewness and concentration. Summary statistics should accompany the chart before making a definitive statement about symmetry.

5.2 Charges by gender

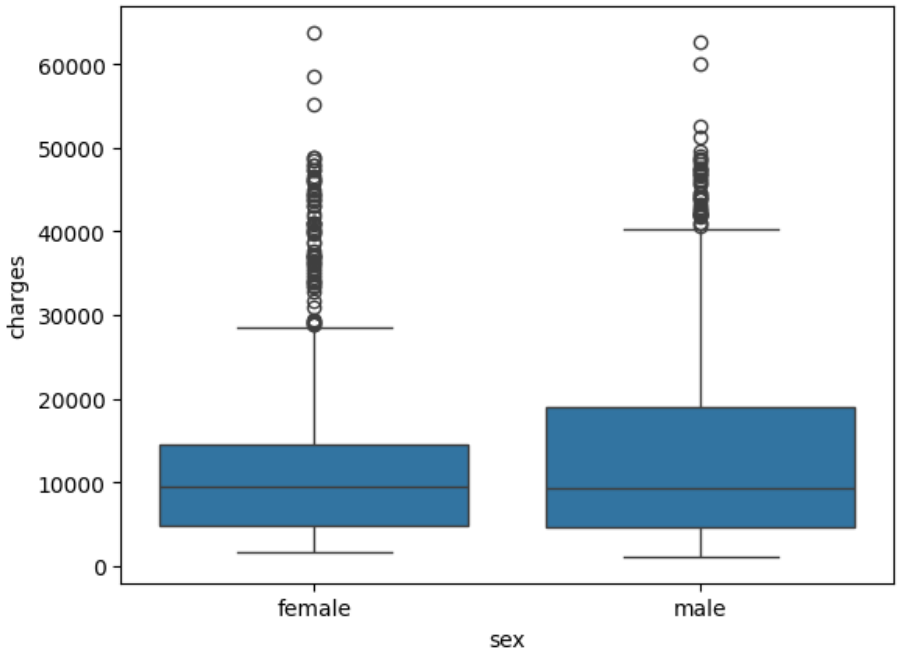


Figure 2. Insurance-charge distributions by gender.

The box plot compares medians, spread and potential outliers. The notebook's decision-printing condition appears reversed and should be corrected before the result is finalized.

5.3 Charges by smoking status

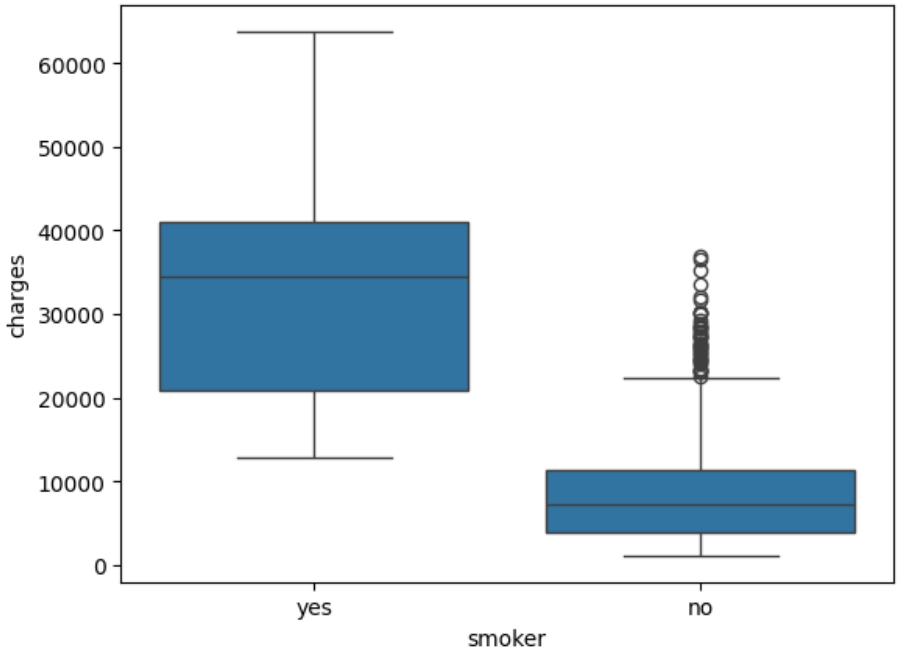


Figure 3. Insurance-charge distributions by smoking status.

The one-sided independent-sample test rejects H■ and concludes that mean charges are higher among smokers than non-smokers.

5.4 Charges across regions

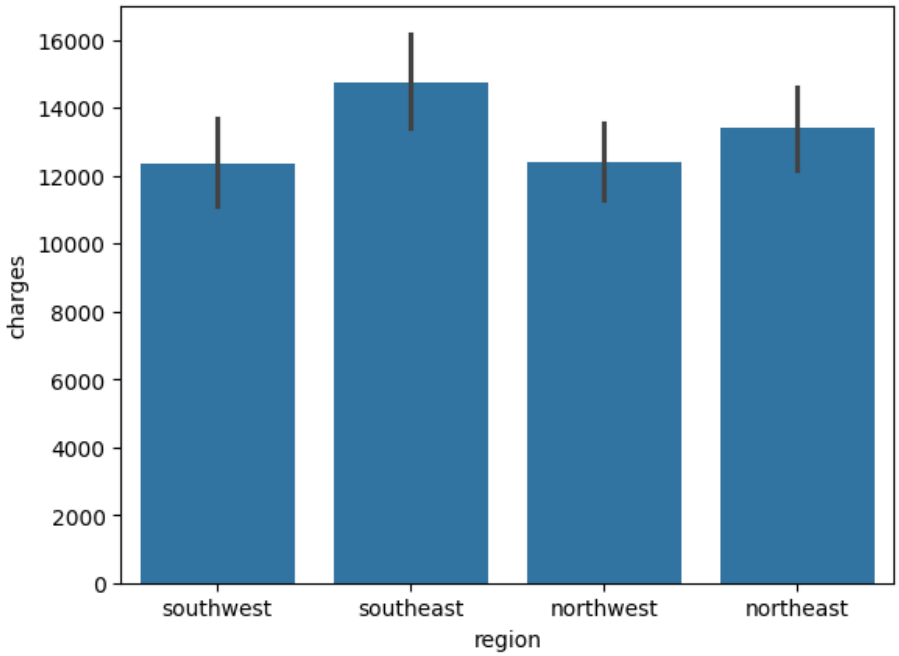


Figure 4. Average insurance charges by region.

The ANOVA rejects H■, indicating at least one regional mean differs. A post-hoc test is needed to identify which pairs differ.

5.5 BMI by gender

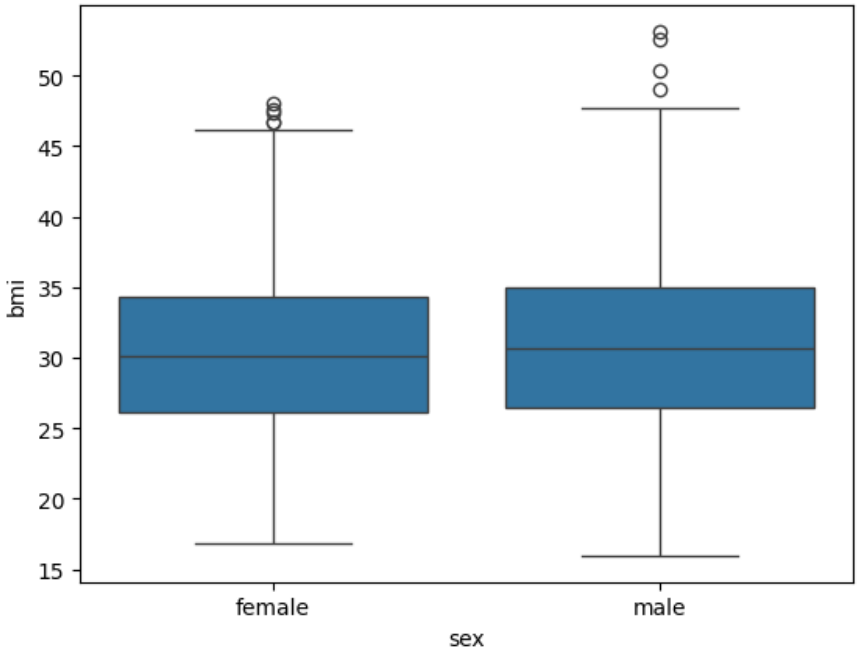


Figure 5. BMI distributions by gender.

The reported p-value is 0.089976, above 0.05; the test fails to reject H■.

5.6 Age and insurance charges

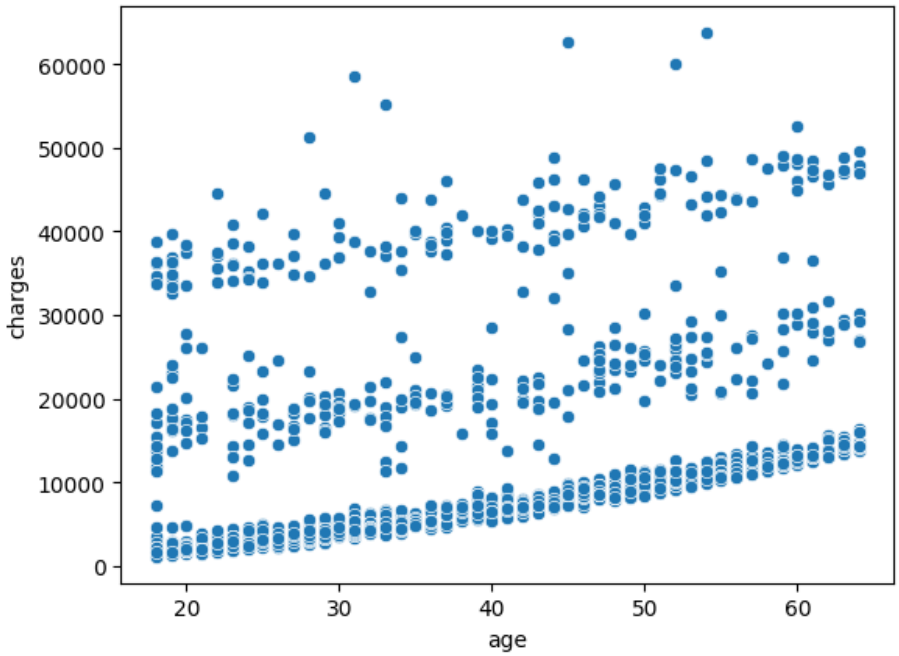


Figure 6. Scatter plot of age against charges.

Pearson's test reports p ≈ 4.89 × 10■²■, evidence of a statistically significant linear association. The coefficient r should be added to show direction and strength.

5.7 Customer count by region

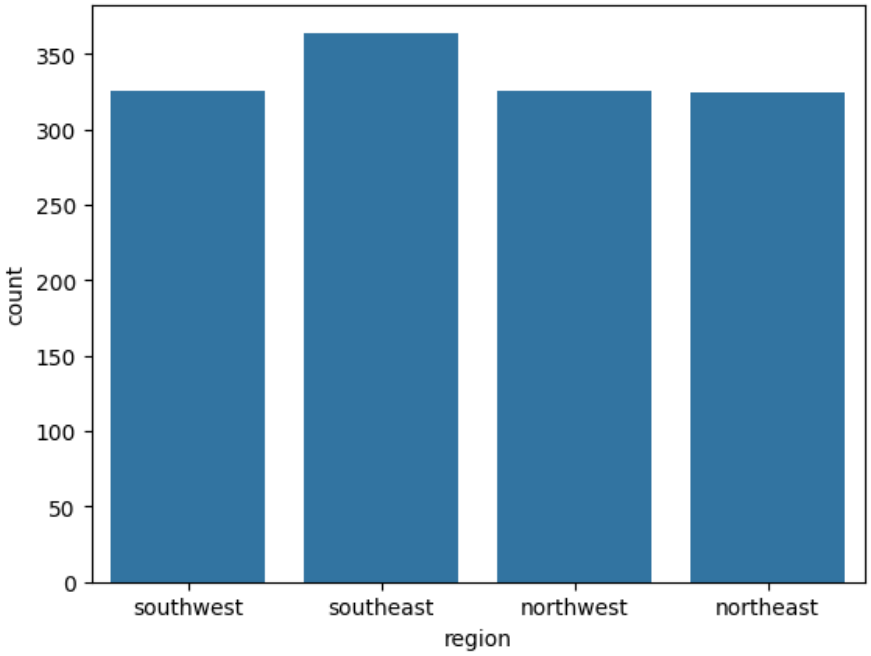


Figure 7. Number of customers by region.

Counts are southeast 364, southwest 325, northwest 325 and northeast 324. The equal expected count is 334.5 per region; p = 0.324562.

5.8 Regional charges by smoking status

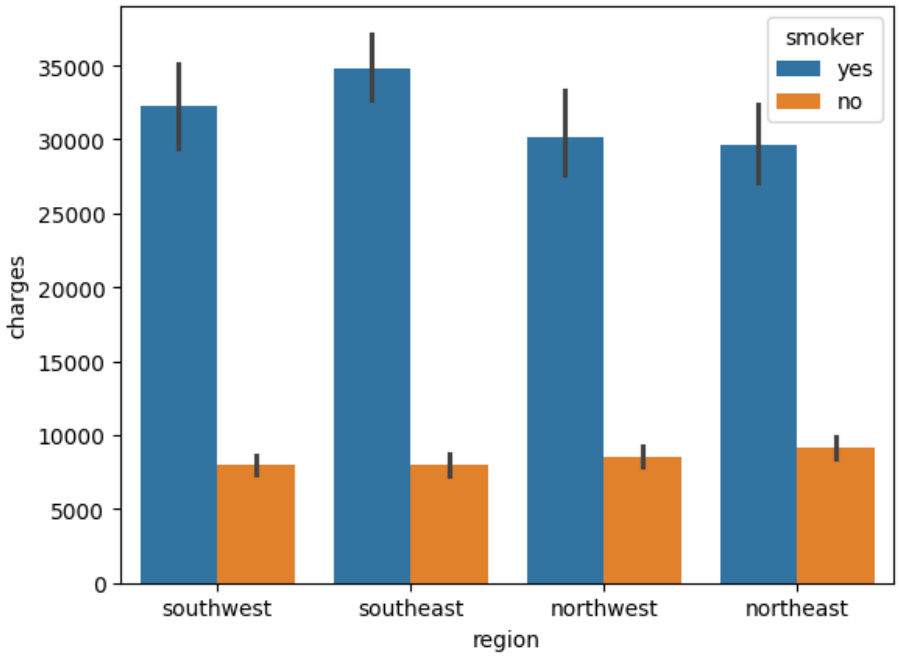


Figure 8. Charges by region and smoking status.

This is not the direct visualization for gender versus smoking status. A 100% stacked bar chart of sex by smoker would align better with that question.

6. Inferential Statistics: Test Register

No.	Question	Test / hypotheses	Recorded result
1	Mean charge vs $13,270	One-sample z-test; H■: μ=13,270; H■: μ≠13,270	Fail to reject H■; exact p-value not shown.
2	Charges by gender	Independent t-test; H■: means equal; H■: means differ	Notebook says fail to reject; code decision condition is reversed.
3	Smoker vs non-smoker charges	One-sided independent t-test; H■: smoker mean higher	Reject H■; smokers have higher mean charges in the tested direction.
4	Charges across regions	One-way ANOVA; H■: all means equal	Reject H■; use post-hoc testing.
5	Gender and smoking status	Nominal association test; H■: independent	Statistic 7.765921; p = 0.005324; reject H■.
6	BMI by gender	Independent t-test; H■: means equal	p = 0.089976; fail to reject H■.
7	Age and charges	Pearson correlation; H■: ρ = 0	p ≈ 4.8867 × 10■²■; reject H■.
8	Equal counts by region	Chi-square goodness-of-fit; H■: equal proportions	p = 0.324562; fail to reject H■.
9	Smoking status and region	Nominal association test; H■: independent	p = 0.061720; fail to reject H■.

7. Key Findings

• Smoking status is associated with higher mean charges in the one-sided comparison.

• Regional mean charges are not all equal according to the recorded ANOVA result.

• The notebook reports an association between gender and smoking status (p = 0.005324).

• The BMI-by-gender test does not reach the 5% significance threshold.

• Age and charges have a significant Pearson relationship, but the correlation coefficient is missing.

• The regional count test and smoking-status/region test are not significant at α = 0.05.

8. Quality Checks and Corrections Before Submission

• Correct the gender-charge decision condition: the notebook currently prints rejection when p > 0.05.

• Fix the regional goodness-of-fit hypotheses: H■ should state equal proportions; H■ should state that at least one differs.

• Confirm whether a z-test is justified; if population standard deviation is unknown, consider a one-sample t-test.

• Prefer Welch's t-test if equal variances cannot be justified.

• Record test statistics, degrees of freedom, exact p-values, group means, sample sizes and confidence intervals.

• For significant ANOVA, add Tukey HSD or another suitable post-hoc test.

• Report Pearson r alongside its p-value.

• Add a direct sex-versus-smoker grouped or 100% stacked bar chart.

• Document dataset source, license, missing values, duplicates and outlier treatment.

9. Limitations

This report uses the supplied notebook's code, markdown and saved outputs. The source CSV was not included, so missing statistics have not been recalculated and no new data-derived charts have been fabricated. Some test outputs are incomplete. The analysis is observational and does not establish causation.

10. Conclusion

The project demonstrates how exploratory visualization and inferential statistics can be combined to study medical insurance charges. Recorded results indicate significant differences by smoking status and region, a gender–smoking association, and an age–charges relationship. Other tests do not provide sufficient evidence to reject their null hypotheses

at the 5% level. Correct the identified logic and hypothesis issues and add complete statistics before final academic submission.

Appendix — Reproducibility

Notebook: Healthcare_project.ipynb
Input expected: insurance.csv
Libraries: pandas, NumPy, Matplotlib, Seaborn, statsmodels and SciPy
Significance level: $\alpha = 0.05$
Run notebook cells in order after setting the correct CSV path.